

S&B: Sutton and Barto, Reinforcement Learning: An Introduction (2nd Edition, 2018): <http://incompleteideas.net/book/RLbook2020.pdf>

Pytorch: https://pytorch.org/tutorials/beginner/deep_learning_60min_blitz.html

A*: <https://www.cs.auckland.ac.nz/courses/compsci709s2c/resources/Mike.d/astarNilsson.pdf>

Q*: <https://prl-theworkshop.github.io/prl2024-icaps/papers/9.pdf>

Atari: <http://klab.tch.harvard.edu/academia/classes/BAI/pdfs/MnihEtAlHassibis15NatureControlDeepRL.pdf>

DeepCubeA: https://cse.sc.edu/~foresta/assets/files/SolvingTheRubiksCubeWithDeepReinforcementLearningAndSearch_Final.pdf

DPG: <http://proceedings.mlr.press/v32/silver14.pdf>

	Subject	Readings	Date	Homework	Quizzes	Project
Week 1	Intro	S&B 1	1/12/26	H0 (Due 1/14)		
	MDPs	S&B 3	1/14/25	H1 (Due 2/3)		
Week 2	Holiday		1/19/26			
	Dynamic Programming	S&B 4	1/21/25			
Week 3	Model-Free RL	S&B 5.1-5.7, 6.1-6.5, 7.1-7.3	1/26/26			
	Deep Neural Networks I		1/28/25		Q1: MDPs and DP	
Week 4	Deep Neural Networks II	PyTorch	2/2/26			
	Deep Q-learning and Value Iteration	Atari and DeepCubeA	2/4/25	H2 (Due 2/24)		
Week 5	Policy Gradients	S&B 13.1-13.8	2/9/26			Proj Proposal (Due 2/25)
	Actor-Critic Methods		2/11/25		Q2: Model-free and Deep RL	
Week 6	Bandits and Exploration	S&B 2	2/16/26			
	Monte Carlo Tree Search	S&B 8.9-8.11	2/18/25			
Week 7	A* and Q* Search	A*, Q*	2/23/26			
	Sparse Reward Problem		2/25/25		Q3: Policy gradient methods	
Week 8	Midterm Review		3/2/26			Proj Progress Rep (Due 3/30)
	Midterm		3/4/25			
Week 9	Holiday		3/9/26			
	Holiday		3/11/25			
Week 10	Model-Based RL and Search	S&B 8.1-8.8	3/16/26	H3 (Due 4/5)		
	Partial observability		3/18/25			
Week 11	Offline RL		3/23/26			
	TRPO and PPO		3/25/25		Q4: Model-based and POMDPs	
Week 12	Multitask RL		3/30/26			
	Explainable RL		4/1/25	H4 (Due 4/21)		
Week 13	Advanced RL and Heuristic Search		4/6/26			
	TBD		4/8/25			
Week 14	TBD		4/13/26			
	Paper Presentations		4/15/25			
Week 15	Paper Presentations		4/20/26			
	Project Presentations		4/22/25			
	Project Presentations		4/27/26			